Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let $F(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$ then $\mathrm{F}(\alpha)\cdot\mathrm{F}(\beta)$ is equal to

(A) $\mathrm{F}(\alpha\beta)$

(B) $\mathrm{F}\left(\frac{\alpha}{\beta}\right)$

(C) $\mathrm{F}(\alpha+\beta)$

(D) $\mathrm{F}(\alpha-\beta)$

Q.2

For each real number x such that $-1<x<1$, let $\mathrm{A}(x)$ be the matrix $(1-x)^{-1}\begin{bmatrix}1 & -x\\ -x & 1\end{bmatrix}$ and $z=\frac{x+y}{1+xy}$. Then

(A) $\mathrm{A}(z)=\mathrm{A}(x)+\mathrm{A}(y)$

(B) $\mathrm{A}(z)=\mathrm{A}(x)[\mathrm{A}(y)]^{-1}$

(C) $\mathrm{A}(z)=\mathrm{A}(x)\mathrm{A}(y)$

(D) $\mathrm{A}(z)=\mathrm{A}(x)-\mathrm{A}(y)$

Q.3

If $A=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$, then A^{16} is equal to :

(A) $\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$

(B) $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$

(C) $\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$

(D) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

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Q.4

If $\begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is square root of identity matrix of order 2 then $-$

(A) $1+\alpha^2+\beta\gamma=0$

(B) $1+\alpha^2-\beta\gamma=0$

(C) $1-\alpha^2+\beta\gamma=0$

(D) $\alpha^2+\beta\gamma=1$

Q.5

Let $\mathrm{A}=\begin{bmatrix} \mathrm{a} & 0 & 0 \\ 0 & \mathrm{a} & 0 \\ 0 & 0 & \mathrm{a} \end{bmatrix}$, then A^{n} is equal to

(A) $\begin{bmatrix} \mathrm{a}^{\mathrm{n}} & 0 & 0 \\ 0 & \mathrm{a}^{\mathrm{n}} & 0 \\ 0 & 0 & \mathrm{a} \end{bmatrix}$

(B) $\begin{bmatrix} \mathrm{a}^{\mathrm{n}} & 0 & 0 \\ 0 & \mathrm{a} & 0 \\ 0 & 0 & \mathrm{a} \end{bmatrix}$

(C) $\begin{bmatrix} a^n & 0 & 0 \\ 0 & a^n & 0 \\ 0 & 0 & a^n \end{bmatrix}$

(D) $\begin{bmatrix} \mathrm{na} & 0 & 0 \\ 0 & \mathrm{na} & 0 \\ 0 & 0 & \mathrm{na} \end{bmatrix}$

Q.6

If $\mathrm{A}=\begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}$ and $\mathrm{B}=\begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$ then which statement is true ?

(A) $\mathrm{AA}^{\mathrm{T}}=\mathrm{I}$

(B) $\mathrm{BB}^{\mathrm{T}}=\mathrm{I}$

(C) $\mathrm{AB}\neq\mathrm{BA}$

(D) $(\mathrm{AB})^{\mathrm{T}}=\mathrm{I}$

Q.7

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If $X = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$, and I is a2×2 identity matrix, then $\mathrm{X}^2 - 2\mathrm{X} + 3\mathrm{I}$ equals to which one of the following?

(A) - I

(B) $-2\mathrm{X}$

(C) $2\mathrm{X}$

(D) $4\mathrm{X}$

Q.8

If $\mathrm{A} = \begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and I is the identity matrix of order 2, then $(\mathrm{I} - \mathrm{A}) \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ is equal to

(A) $I + A$

(B) $I - A$

(C) $\mathrm{A} - \mathrm{I}$

(D) A

Q.9

Matrix A such that $\mathrm{A}^2 = 2\,\mathrm{A} - \mathrm{I}$, where I is the identity matrix, then for $\mathrm{n} \geq 2,\ \mathrm{A}^{\mathrm{n}}$ is equal to

(A) $2^{\mathrm{n}-1}\,\mathrm{A} - (\mathrm{n}-1)\mathrm{I}$

(B) $2^{\mathrm{n}-1}\,\mathrm{A} - \mathrm{I}$

(C) $\mathrm{nA} - (\mathrm{n}-1)\mathrm{I}$

(D) $\mathrm{nA} - \mathrm{I}$

Q.10

Consider the matrices

$$A = \begin{bmatrix} 4 & 6 & -1 \\ 3 & 0 & 2 \\ 1 & -2 & 5 \end{bmatrix}, B = \begin{bmatrix} 2 & 4 \\ 0 & 1 \\ -1 & 2 \end{bmatrix}, C = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Out of the given matrix products, which one is not defined.

(A) $(\mathrm{AB})^{\mathrm{T}}\mathrm{C}$

(B) $\mathrm{C}^{\mathrm{T}}\mathrm{C}(\mathrm{AB})^{\mathrm{T}}$

(C) $\mathrm{C}^{\mathrm{T}}\mathrm{AB}$

(D) $\mathrm{A}^{\mathrm{T}}\mathrm{ABB}^{\mathrm{T}}\mathrm{C}$

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Q.11

Let $\mathrm{A} = \begin{bmatrix} 5 & 6 & 1 \\ 2 & -1 & 5 \end{bmatrix}$. Let there exist a matrix B such that $AB = \begin{bmatrix} 35 & 49 \\ 29 & 13 \end{bmatrix}$. What is B equal to?

(A) $\begin{bmatrix} 5 & 1 & 4 \\ 2 & 6 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & 6 & 3 \\ 5 & 1 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$

(D) $\begin{bmatrix} 2 & 5 \\ 6 & 1 \\ 3 & 4 \end{bmatrix}$

Q.12

Let $\mathrm{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ be a square matrix of order 3.
Then for any positive integer n, what is A^{n} equal to?

(A) A

(B) $3^{\mathrm{n}}\mathrm{A}$

(C) $\left(3^{n-1}\right) A$

(D) $3\ \mathrm{A}$

Q.13

If $\mathrm{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, then what is A^{n} equal to ?

(A) $\begin{bmatrix} 2^{\mathrm{n}} & 2^{\mathrm{n}} \\ 2^{\mathrm{n}} & 2^{\mathrm{n}} \end{bmatrix}$

(B) $\begin{bmatrix} 2\mathrm{n} & 2\mathrm{n} \\ 2\mathrm{n} & 2\mathrm{n} \end{bmatrix}$

(C) $\begin{bmatrix} 2^{2\mathrm{n}-1} & 2^{2\mathrm{n}-1} \\ 2^{2\mathrm{n}-1} & 2^{2\mathrm{n}-1} \end{bmatrix}$

(D) $\begin{bmatrix} 2^{2n+1} & 2^{2n+1} \\ 2^{2n+1} & 2^{2n+1} \end{bmatrix}$

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Q.14

If $\mathrm{A}=\begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix}$ is a 2×2 matrix and $\mathrm{f}(\mathrm{x})=\mathrm{x}^2-\mathrm{x}+2$ is a polynomial, then what is $\mathrm{f}(\mathrm{A})$?

(A) $\begin{bmatrix}1 & 7\\ 1 & 7\end{bmatrix}$

(B) $\begin{bmatrix}2 & 6\\ 0 & 8\end{bmatrix}$

(C) $\begin{bmatrix}2 & 6\\ 0 & 6\end{bmatrix}$

(D) $\begin{bmatrix}2 & 6\\ 0 & 7\end{bmatrix}$

Q.15

If $A=\begin{pmatrix}p & q\\ 0 & 1\end{pmatrix}$, then $A^8=\begin{pmatrix}p^8 & q\left(\frac{p^8-1}{p-1}\right)\\ 0 & K\end{pmatrix}$. The value of k is

(A) 1

(B) 0

(C) 2

(D) -1

Q.16

$\mathrm{A}=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ then A^{100} :

(A) $2^{100}\ \mathrm{A}$

(B) $2^{99}\ \mathrm{A}$

(C) $2^{101}\ \mathrm{A}$

(D) None of the above

Q.17

If A and B are two matrices such that $\mathrm{AB}=\mathrm{B}$ and $\mathrm{BA}=\mathrm{A}$, then $\mathrm{A}^2+\mathrm{B}^2$ is equal to

(A) $2\mathrm{AB}$

(B) $2\mathrm{BA}$

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(C) $\mathrm{A}+\mathrm{B}$

(D) AB

Q.18

If $B=\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C=\begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ and $X=BC$, find $\mathrm{X^n}$

(A) 0

(B) I

(C) $2\mathrm{I}$

(D) None of these

Q.19

If $\mathrm{A}=\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ then $\lim_{n\to\infty}\frac{1}{n}A^n$ is

(A) a null matrix

(B) an identity matrix

(C) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

(D) None of these

Q.20

Let $\mathrm{A,B,C,D}$ be (not necessarily square) real matrices such that $\mathrm{A^T=BCD;B^T=CDA;}$ $\mathrm{C^T=DAB}$ and $\mathrm{D^T=ABC}$ forthematrix $\mathrm{S=ABCD,S^3=}$

(A) I

(B) $\mathrm{S^2}$

(C) S

(D) O

Answers & Solutions

Q.1 Answer:

$\mathrm{F}(\alpha+\beta)$

Solution:

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$$\mathrm{F}(\alpha)\cdot\mathrm{F}(\beta)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\beta & -\sin\beta & 0\\ \sin\beta & \cos\beta & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$F(\alpha)\cdot\mathrm{F}(\beta)=\begin{bmatrix}\cos(\alpha+\beta) & -\sin(\alpha+\beta) & 0\\ \sin(\alpha+\beta) & \cos(\alpha+\beta) & 0\\ 0 & 0 & 1\end{bmatrix}=F(\alpha+\beta)$$

Q.2 Answer:

$$\mathrm{A}(z)=\mathrm{A}(x)\mathrm{A}(y)$$

Solution:

$$\mathrm{A}(z)=\mathrm{A}\left(\frac{x+y}{1+xy}\right)=\left[\frac{1+xy}{(1-x)(1-y)}\right]$$
$$\begin{bmatrix}1 & -\left(\frac{x+y}{1+xy}\right)\\ -\left(\frac{x+y}{1+xy}\right) & 1\end{bmatrix}$$
$$\therefore\quad \mathrm{A}(x)\cdot\mathrm{A}(y)=\mathrm{A}(z)$$

Q.3 Answer:

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Solution:

We have $A=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$

Now, $A^2=A\cdot A=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}=-I$

where $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ **is identity matrix** $\left(A^2\right)^8=(-I)^8=I$

Hence, $A^{16}=I$

Q.4 Answer:

$$\alpha^2+\beta\gamma=1$$

Solution:

$$\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}=\sqrt{I_2};\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$
$$\Rightarrow \alpha^2+\beta\gamma=1$$

Q.5 Answer:

$$\begin{bmatrix}a^n & 0 & 0\\ 0 & a^n & 0\\ 0 & 0 & a^n\end{bmatrix}$$

Solution:

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$$A^2 = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & a^2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & a^2 \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} = \begin{bmatrix} a^3 & 0 & 0 \\ 0 & a^3 & 0 \\ 0 & 0 & a^3 \end{bmatrix}$$

Q.6 Answer:

$(AB)^T = I$

Solution:

Here $AA^T = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}\begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$(BB^T)_{11} = (d)^2 + (a)^2 \neq 1$

$(AB)_{11} = 8 - 7 = 1, (BA)_{11} = 8 - 7 = 1$

$\therefore AB \neq BA$ **may be not true**

$\text{Now } AB = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$

$= \begin{pmatrix} 8-7 & 2-2 \\ -28+28 & -7+8 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; (AB)^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

Q.7 Answer:

$2X$

Solution:

Given matrix is:

$X = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$

$\therefore X^2 = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$

$= \begin{bmatrix} 1 & -2-6 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & -8 \\ 0 & 9 \end{bmatrix}$

So, the given expression is:

$X^2 - 2X + 3I = \begin{bmatrix} 1 & -8 \\ 0 & 9 \end{bmatrix} - 2\begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & -8 \\ 0 & 9 \end{bmatrix} + \begin{bmatrix} -2 & +4 \\ 0 & -6 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

$= \begin{bmatrix} 1-2+3 & -8+4 \\ 0 & 9-6+3 \end{bmatrix}$

$= \begin{bmatrix} 2 & -4 \\ 0 & 6 \end{bmatrix} = 2\begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} = 2X$

Q.8 Answer:

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$I + A$

Solution:

Here, $\mathrm{A} = \begin{bmatrix} 0 & -\mathrm{t} \\ \mathrm{t} & 0 \end{bmatrix}$, **where** $\mathrm{t} = \tan\left(\frac{\alpha}{2}\right)$

Now, $\cos\alpha = \frac{1-\tan^2\left(\frac{\alpha}{2}\right)}{1+\tan^2\left(\frac{\alpha}{2}\right)} = \frac{1-t^2}{1+t^2}$

and $\sin\alpha = \frac{2\tan\left(\frac{\alpha}{2}\right)}{1+\tan^2\left(\frac{\alpha}{2}\right)} = \frac{2t}{1+t^2}$

$$= (\mathrm{I} - \mathrm{A}) \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -t \\ +t & 0 \end{bmatrix} \right) \begin{bmatrix} \frac{1-t^2}{1+t^2} & \frac{-2t}{1+t^2} \\ \frac{2t}{1+t^2} & \frac{1-t^2}{1+t^2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix} \begin{bmatrix} \frac{1-t^2}{1+t^2} & \frac{-2t}{1+t^2} \\ \frac{2t}{1+t^2} & \frac{1-t^2}{1+t^2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1-t^2+2t^2}{1+t^2} & \frac{-2t+t\left(1-t^2\right)}{1+t^2} \\ \frac{-t\left(1-t^2\right)+2t}{1+t^2} & \frac{2t^2+1-t^2}{1+t^2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1+t^2}{1+t^2} & \frac{-2t+t-t^3}{1+t^2} \\ \frac{-t+t^3+2t}{1+t^2} & \frac{2t^2+1-t^2}{1+t^2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1+t^2}{1+t^2} & \frac{-t\left(1+t^2\right)}{1+t^2} \\ \frac{t\left(1+t^2\right)}{1+t^2} & \frac{1+t^2}{1+t^2} \end{bmatrix} = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix}$$

Also, $\mathrm{I} + \mathrm{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -\mathrm{t} \\ \mathrm{t} & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & -\mathrm{t}+0 \\ \mathrm{t}+0 & 0+1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix} = (I - A) \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

Q.9 Answer:

$\mathrm{nA} - (\mathrm{n}-1)\mathrm{I}$

Solution:

Given $\mathrm{A}^2 = 2\,\mathrm{A} - \mathrm{I}$

Now $A^3 = A\left(A^2\right) = A(2A - I)$

$= 2\,\mathrm{A}^2 - \mathrm{A} = 2(2\,\mathrm{A} - \mathrm{I}) - \mathrm{A} = 3\,\mathrm{A} - 2\mathrm{I}$

$A^4 = A\left(A^3\right) = A(3A - 2I)$

$= 3\,\mathrm{A}^2 - 2\,\mathrm{A} = 3(2\,\mathrm{A} - \mathrm{I}) - 2\,\mathrm{A} = 4\,\mathrm{A} - 3\mathrm{I}$

Following this, we can say $\mathrm{A}^{\mathrm{n}} = \mathrm{nA} - (\mathrm{n}-1)\mathrm{I}$

Q.10 Answer:

$\mathrm{C}^{\mathrm{T}}\mathrm{C}(\mathrm{AB})^{\mathrm{T}}$

Solution:

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$A \to 3 \times 3,\ B \to 3 \times 2, C \to 3 \times 1$
$AB \to 3 \times 2 \Rightarrow (AB)^T = 2 \times 3 \Rightarrow (AB)^T C$ **is**
defined $\Rightarrow \quad C^T \to 1 \times 3, \Rightarrow C^T C \to 1 \times 1$
Hence $C^T C (AB)^T$ **is not defined.Now,** $C^T AB$ **is**
also defined. $A^T \to 3 \times 3,\ B^T \to 2 \times 3; \quad A^T A \to 3 \times 3$
$BB^T \to 3 \times 3 \Rightarrow A^T ABB^T \to 3 \times 3$
$\Rightarrow \quad A^T ABB^T C$ **is defined**

Q.11 Answer:

$$\begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$$

Solution:

$A = \begin{bmatrix} 5 & 6 & 1 \\ 2 & -1 & 5 \end{bmatrix}$ **and let** $B = \begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$

$$\therefore \quad AB = \begin{bmatrix} 5 & 6 & 1 \\ 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 25+6+4 & 10+36+3 \\ 10-1+20 & 4-6+15 \end{bmatrix}$$

$$= \begin{bmatrix} 35 & 49 \\ 29 & 13 \end{bmatrix}$$

Q.12 Answer:

$\left(3^{n-1}\right) A$

Solution:

Given matrix is $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$\text{So } A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\text{So } = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\text{and } A^3 = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Similarly $A^4 = 3^3$ A. Hence, $A^n = 3^{n-1}$ A

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Q.13 Answer:

$\begin{bmatrix} 2^{2n-1} & 2^{2n-1} \\ 2^{2n-1} & 2^{2n-1} \end{bmatrix}$

Solution:

Given matrix is:

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 4+4 & 4+4 \\ 4+4 & 4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 2^3 & 2^3 \\ 2^3 & 2^3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 16+16 & 16+16 \\ 16+16 & 16+16 \end{bmatrix}$$

$$= \begin{bmatrix} 32 & 32 \\ 32 & 32 \end{bmatrix} = \begin{bmatrix} 2^5 & 2^5 \\ 2^5 & 2^5 \end{bmatrix}$$

Going this way we get $A^4 = \begin{bmatrix} 2^7 & 2^7 \\ 2^7 & 2^7 \end{bmatrix}$

$$\Rightarrow A^n = \begin{bmatrix} 2^{2n-1} & 2^{2n-1} \\ 2^{2n-1} & 2^{2n-1} \end{bmatrix}$$

Q.14 Answer:

$\begin{bmatrix} 2 & 6 \\ 0 & 8 \end{bmatrix}$

Solution:

Given that, $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2+6 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}$$

Since, $f(x) = x^2 - x + 2$

Putting A in place of x $f = A^2 - A + 2I$

$$= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1+2 & 8-2+0 \\ 0-0+0 & 9-3+2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 6 \\ 0 & 8 \end{bmatrix}$$

Q.15 Answer:

1

Solution:

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$$A^2 = \begin{pmatrix} p & q \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & q \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p^2 & pq+q \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} p^2 & q(p+1) \\ 0 & 1 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} p & q \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p^2 & pq+q \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} p^3 & q\left(p^2+p+1\right) \\ 0 & 1 \end{pmatrix}$$

Similarly, $A^4 = \begin{pmatrix} p^4 & q\left(p^3+p^2+p+1\right) \\ 0 & 1 \end{pmatrix}$ **and so on.**

$$A^8 = \begin{pmatrix} p^8 & q\left(p^7+p^8+\ldots+1\right) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p^8 & q\left(\frac{p^8-1}{p-1}\right) \\ 0 & 1 \end{pmatrix}$$

Q.16 Answer:

$2^{99}\ \mathrm{A}$

Solution:

Let $\mathrm{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$\mathrm{A}^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 2\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 2\ \mathrm{A}$$

$$\mathrm{A}^3 = 2^2\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \mathrm{A}^4 = 2^3\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\mathrm{A}^3 = 2^2\ \mathrm{A}$$

$$\mathrm{A}^4 = 2^3\ \mathrm{A} \quad \therefore \quad \mathrm{A}^n = 2^{n-1}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow A^{100} = 2^{100-1}A \therefore A^{100} = 2^{99}A$$

Q.17 Answer:

$\mathrm{A}+\mathrm{B}$

Solution:

We have, $\mathrm{A}^2+\mathrm{B}^2 = \mathrm{AA}+\mathrm{BB} = \mathrm{A}(\mathrm{BA})+\mathrm{B}(\mathrm{AB})$

$(\therefore \mathrm{AB} = \mathrm{B}$ **and** $\mathrm{BA}+\mathrm{A})$

$= (\mathrm{AB})\mathrm{A}+(\mathrm{BA})\mathrm{B}$

$= \mathrm{BA}+\mathrm{AB} = \mathrm{A}+\mathrm{B} \quad (\therefore \mathrm{AB} = \mathrm{B}$ **and** $\mathrm{BA} = \mathrm{A})$

Q.18 Answer:

I

Solution:

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$$B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$$
$$X = BC$$
$$= \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$$
$$\Rightarrow X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\Rightarrow \quad X = I$$
$$\therefore \quad X^n = I$$

Q.19 Answer:

a null matrix

Solution:

$$A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$$
$$\frac{1}{\mathrm{n}}\mathrm{A}^{\mathrm{n}} = \begin{bmatrix} \frac{\cos \mathrm{n}\theta}{\mathrm{n}} & \frac{\sin \mathrm{n}\theta}{\mathrm{n}} \\ -\frac{\sin \mathrm{n}\theta}{\mathrm{n}} & \frac{\cos \mathrm{n}\theta}{\mathrm{n}} \end{bmatrix}$$
But $-1 \leq \cos n\theta \leq 1$ **and** $-1 \leq \sin n\theta \leq 1$
$\lim_{n\to\infty} \frac{\sin n\theta}{n} = 0, \lim_{n\to\infty} \frac{\cos n\theta}{n} = 0$
$$\lim_{n\to\infty} \frac{1}{n}\, \mathrm{A}^n = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Q.20 Answer:

S

Solution:

$$\mathrm{S} = \mathrm{ABCD} = \mathrm{A(BCD)} = \mathrm{AA}^{\mathrm{T}}$$
$$\mathrm{S}^3 = (\mathrm{ABCD})(\mathrm{ABCD})(\mathrm{ABCD})$$
$$= (\mathrm{ABC})(\mathrm{DAB})(\mathrm{CDA})(\mathrm{BCD})$$
$$= \mathrm{D}^{\mathrm{T}}\mathrm{C}^{\mathrm{T}}\mathrm{B}^{\mathrm{T}}\mathrm{A}^{\mathrm{T}}$$
$$= (\mathrm{BCD})^{\mathrm{T}}\mathrm{A}^{\mathrm{T}} = \mathrm{AA}^{\mathrm{T}}$$
$$\text{From (1) and (2)}, \mathrm{S} = \mathrm{S}^3$$

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